

JavaScript Interview Cheat Sheet

1. Event Loop

- Call stack, microtask queue, macrotask queue.
- Microtasks run before macrotasks.

2. Closures

- Inner function remembering outer scope variables.
- Used for data privacy, factories, debouncing.

3. Prototype Chain

- Objects inherit via prototype chain.

4. == vs ===

- == does type coercion; === strict comparison.

5. Hoisting

- var hoisted as undefined; let/const hoisted but in TDZ.

6. Pass by Value / Reference

- Primitives by value; objects by reference copy.

7. Debounce / Throttle

- Debounce: wait after last call.
- Throttle: run once per interval.

8. this in JS

- Depends on call-site; arrow functions use lexical this.

9. Map, Set, WeakMap, WeakSet

- Map/Set strong refs; WeakMap/WeakSet garbage-collected keys.

10. Currying

- Transform $f(a,b,c) \rightarrow f(a)(b)(c)$.

11. Memoization

- Cache function outputs.

12. Deep vs Shallow Copy

- Shallow copies top level; deep copies all levels.

13. Event Delegation

- Attach listener to parent; rely on bubbling.

14. Polyfills

- Custom map, filter, reduce.

15. Promises

- Pending → fulfilled/rejected; immutable.

16. Promise.all / race

- all waits for all to resolve; race returns first settled.

17. Async/Await

- Syntax sugar over promises; uses microtask queue.

18. Generators

- Functions paused via yield.

19. V8 Internals

- Parser → bytecode → optimized machine code.

20. Single-threaded JS

- DOM safety; async via Web APIs.

21. TDZ

- let/const inaccessible before init.

22. Optional Chaining / Nullish

- obj?.a; x ?? y returns y only if x is null/undefined.

23. Modules

- CommonJS: require; ESM: import/export.

24. Immutable vs Mutable

- Primitives immutable; objects mutable.

25. Tail Call Optimization

- Reuse stack frame for tail-position calls.